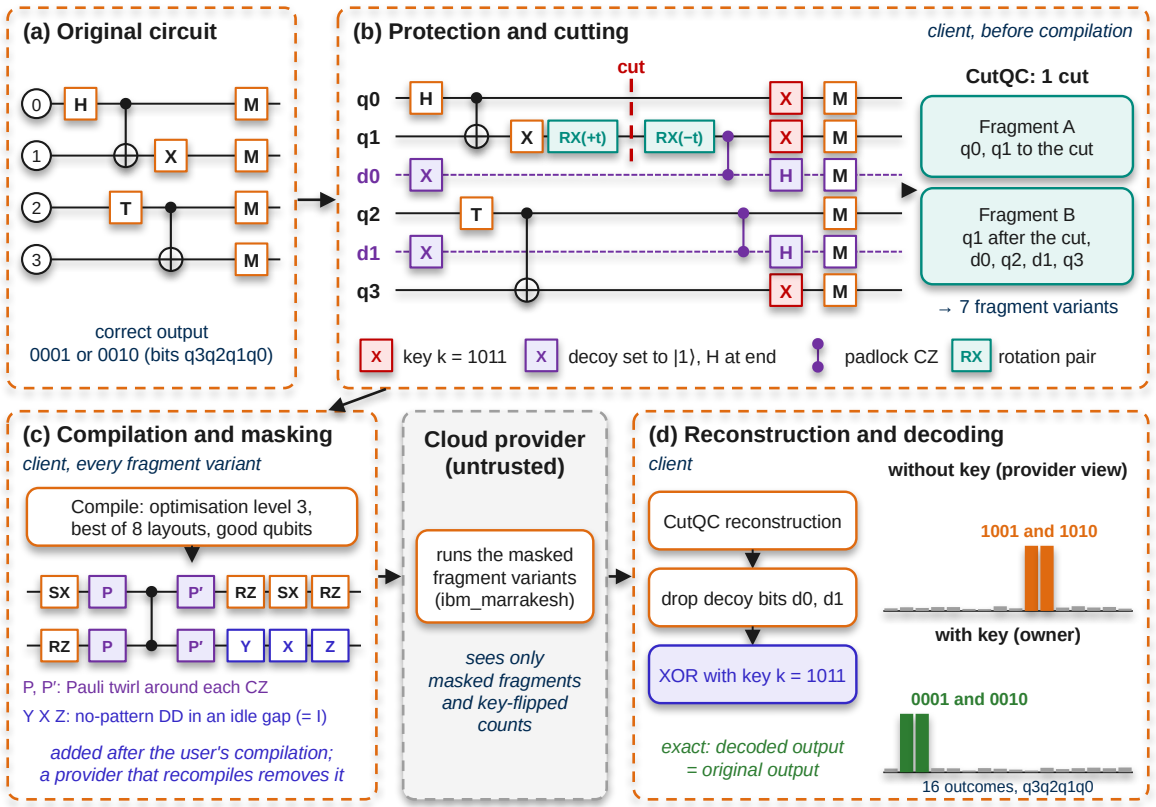

(a) Original circuit
0
1
2
3
H
X
T
M
correct output
0001 or 0010 (bits q3q2q1q0)
(b) Protection and cutting
client, before compilation
cut
q0
q1
d0
q2
d1
q3
RX(+t)
RX(−t)
CutQC: 1 cut
Fragment A
q0, q1 to the cut
Fragment B
q1 after the cut,
d0, q2, d1, q3
→ 7 fragment variants
key k = 1011
decoy set to |1⟩, H at end
padlock CZ
RX
rotation pair
(c) Compilation and masking
client, every fragment variant
Compile: optimisation level 3,
best of 8 layouts, good qubits
SX
P
P′
RZ
Y
P, P′: Pauli twirl around each CZ
Y X Z: no-pattern DD in an idle gap (= I)
added after the user's compilation;
a provider that recompiles removes it
Cloud provider
(untrusted)
runs the masked
fragment variants
(ibm_marrakesh)
sees only
masked fragments
and key-flipped
counts
(d) Reconstruction and decoding
client
CutQC reconstruction
drop decoy bits d0, d1
XOR with key k = 1011
exact: decoded output
= original output
without key (provider view)
1001 and 1010
with key (owner)
0001 and 0010
16 outcomes, q3q2q1q0